THE TWO-SIDED IDEAL GENERATED BY THE COMMUTATORS OF THE ASKEY-WILSON ALGEBRA COINCIDES WITH THE KERNEL OF ITS POLYNOMIAL QUOTIENT

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Allowed standard prerequisites (editorial).

- The homomorphism from the universal Askey-Wilson algebra onto the polynomial algebra, and its surjectivity. Same source, Lemma labelled lem:com2; invoked in the counted statement at line 2806 and inside the counted proof at lines 2811-2818. Supplies the map whose kernel Gamma is one of the two objects compared in the statement.
- The monomials in the barred generators form a basis of the polynomial algebra. Used inside the counted proof at lines 2835-2847. Gives that the monomials A^i B^j C^k form a complement M of Gamma in the algebra, so that the sum Delta = M + Gamma is direct.

Source reference. Original proposition 1.54: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 2806–2806. Navigation pointer for \ref{lem:com2}

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used inside the counted span (line 2806); the argument that consumes it is retained in the span.

Proposition 0.1. *The following coincide:*

(*i*) *The* 2*-sided ideal* $\Delta[\Delta,\Delta]\Delta$*;*

(*ii*) *The kernel of the homomorphism* $\Delta \to \mathbb{F}[\overline{A},\overline{B},\overline{C}]$ *from Lemma* 1.54*.*

Proof. Let Γ denote the kernel of the homomorphism $\Delta \to \mathbb{F}[\overline{A},\overline{B},\overline{C}]$ and note that Γ is a 2-sided ideal of Δ. We have $[\Delta,\Delta] \subseteq \Gamma$ since $\mathbb{F}[\overline{A},\overline{B},\overline{C}]$ is commutative, and $\Delta[\Delta,\Delta]\Delta \subseteq \Gamma$ since Γ is a 2-sided ideal of Δ. The elements $\{\overline{A}^i\overline{B}^j\overline{C}^k | i,j,k \geq 0\}$ form a basis for the $\mathbb{F}$-vector space $\mathbb{F}[\overline{A},\overline{B},\overline{C}]$. Therefore the elements $\{A^iB^jC^k | i,j,k \geq 0\}$ form a basis for a complement of Γ in Δ. Denote this complement by M, so the sum $\Delta = M + \Gamma$ is direct. We now show $\Delta = M + \Delta[\Delta,\Delta]\Delta$. The $\mathbb{F}$-algebra Δ is generated by A, B, C. Therefore the $\mathbb{F}$-vector space Δ is spanned by elements of the form $x_1x_2\cdots x_n$ where $n \geq 0$ and $x_i \in \{A,B,C\}$ for $1 \leq i \leq n$. Let $x_1x_2\cdots x_n$ denote such an element. Then $\Delta[\Delta,\Delta]\Delta$ contains

$$x_1\cdots x_{i-1}x_i\cdots x_n - x_1\cdots x_ix_{i-1}\cdots x_n$$

for $2 \leq i \leq n$. Therefore $\Delta[\Delta,\Delta]\Delta$ contains

$$x_1x_2\cdots x_n - A^iB^jC^k,$$

where i, j, k denote the number of times A, B, C appear among $x_1, x_2, \ldots, x_n$. Observe $A^iB^jC^k \in M$ so $x_1x_2\cdots x_n \in M + \Delta[\Delta,\Delta]\Delta$. Therefore $\Delta = M + \Delta[\Delta,\Delta]\Delta$. We already showed $\Delta[\Delta,\Delta]\Delta \subseteq \Gamma$ and the sum $\Delta = M + \Gamma$ is direct. By these comments $\Delta[\Delta,\Delta]\Delta = \Gamma$. □

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.